SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*
Assessment Tool Weightage: 40%
SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4 **Total Marks:40**

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS)**.** Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer**.** Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer**.** Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT)**.** Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

EXPERIMENT 1

QUESTION 1
Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

ANSWER

Aim

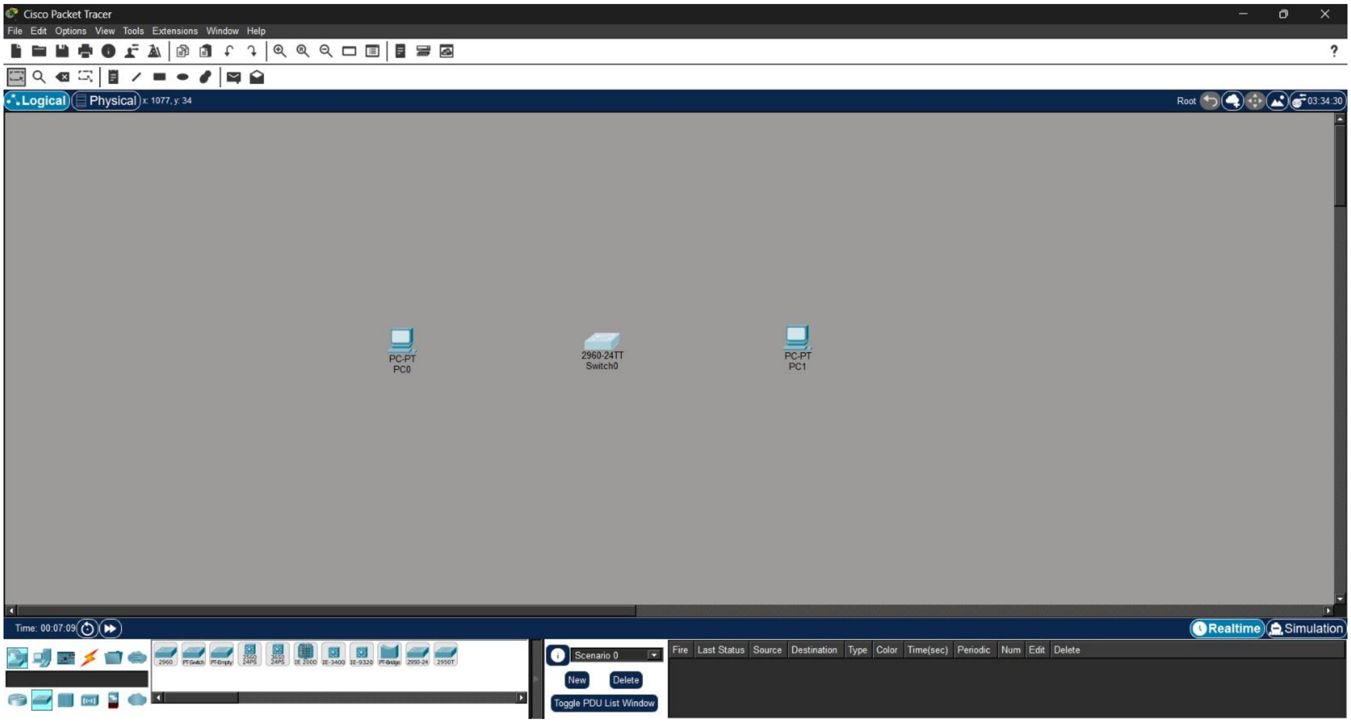
To design a simple LAN consisting of two PCs connected through a switch in Cisco Packet Tracer, configure IP addressing on both hosts, verify connectivity using the ping command, capture the packet exchange, and analyze the Ethernet frame fields — Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS) — in order to understand how the Data Link layer uses MAC addressing to ensure successful frame delivery within a local network.

Procedure

1. Open Cisco Packet Tracer and create a new blank project.
2. Place two End Devices (PC0 and PC1) and one 2960 Switch onto the workspace.
3. Connect PC0 and PC1 to the switch using Copper Straight-Through cables and confirm the link lights turn green.
4. Configure the IPv4 address and subnet mask on PC0 (e.g., 192.168.1.1 / 255.255.255.0) through Desktop → IP Configuration.
5. Configure the IPv4 address and subnet mask on PC1 (e.g., 192.168.1.2 / 255.255.255.0) in the same way.
6. Open the Command Prompt on PC0 and run the ping command to PC1's IP address to verify end-to-end connectivity.
7. Switch Packet Tracer to Simulation Mode and set the Edit Filters to capture only ICMP traffic.
8. Re-run the ping and use Auto Capture / Play to capture the frame exchanged between PC0 and PC1.
9. Open the PDU Information window for the captured packet and inspect the OSI Model → Layer 2 tab to view the Ethernet II header fields.
10. Record the Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS) from the captured frame, and analyze how they enabled successful local delivery.

Step-by-Step Implementation

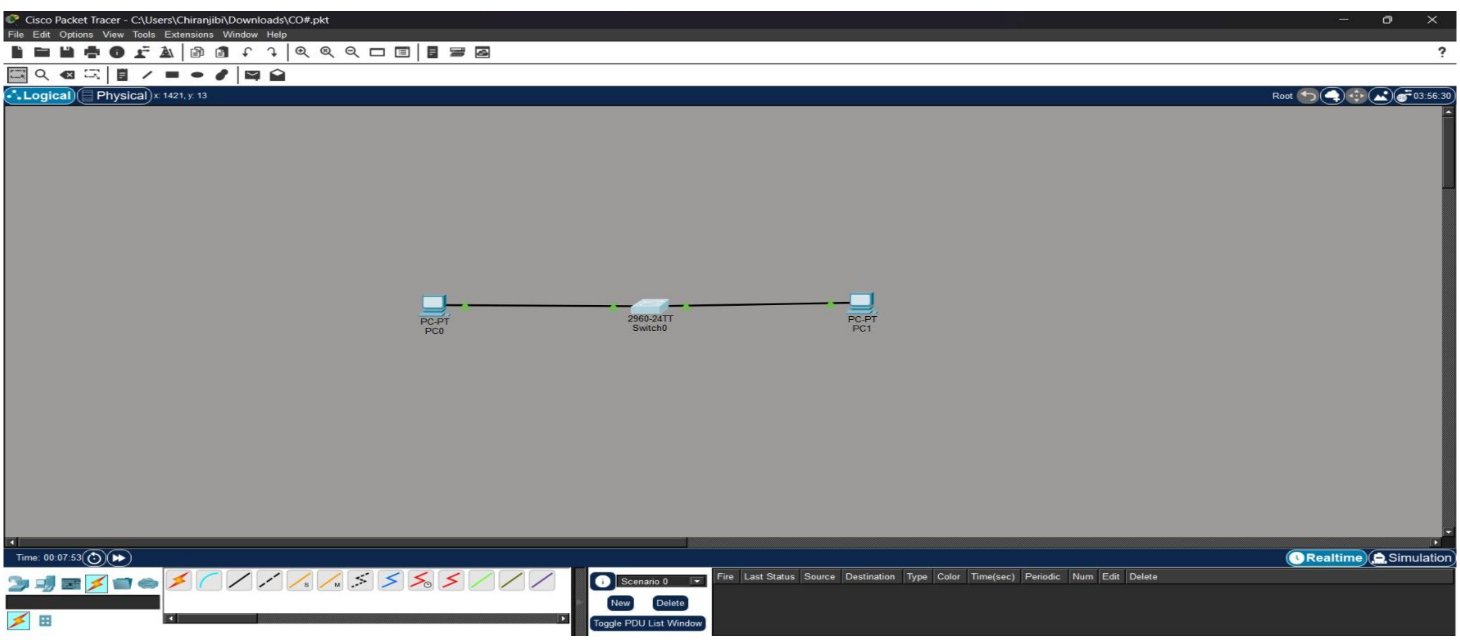
Step 1: Create the Topology



Explanation:

PC0 and PC1 were dragged onto the workspace from the End Devices category, and a 2960 Switch was placed between them from the Switches category. This forms the physical layout needed for a simple two-host LAN.

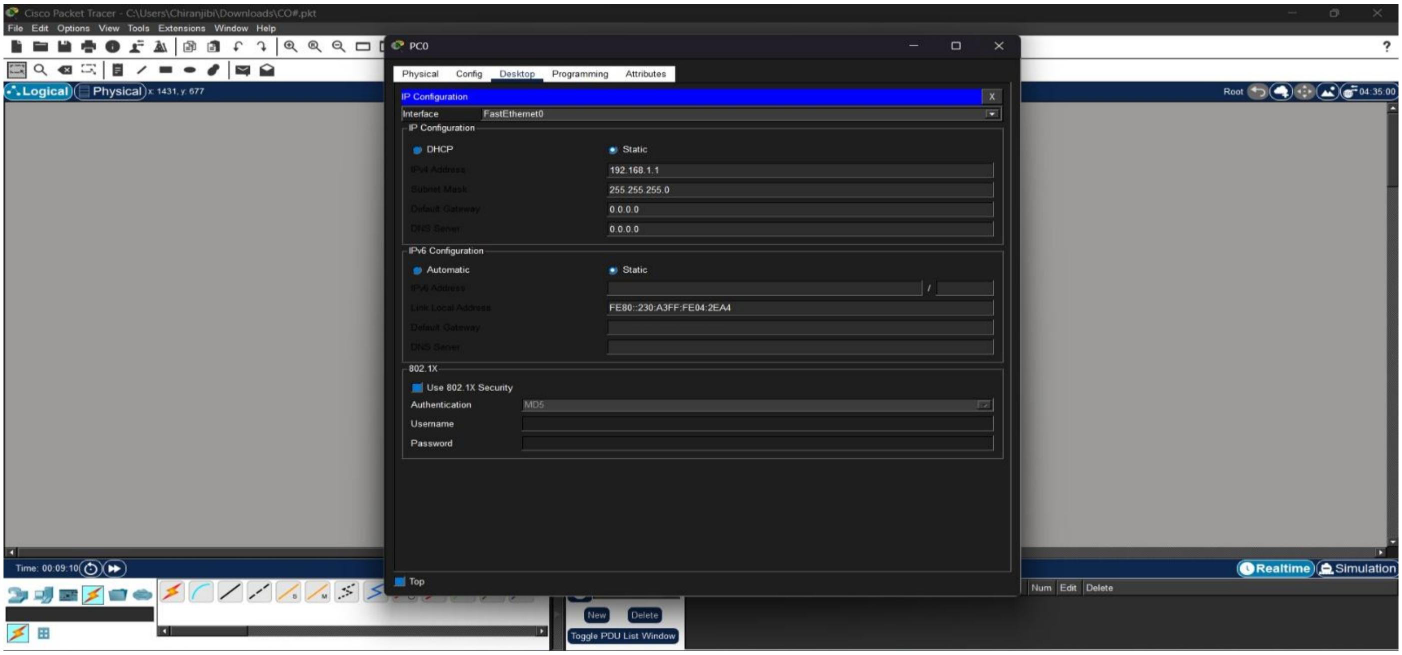
Step 2: Cable the Devices



Explanation:

A Copper Straight-Through cable was used to connect PC0's FastEthernet0 port to the switch's FastEthernet0/1 port, and another cable was used to connect PC1's FastEthernet0 port to FastEthernet0/2. The green dots at both ends of each cable confirm that the physical (Layer 1) connection is active.

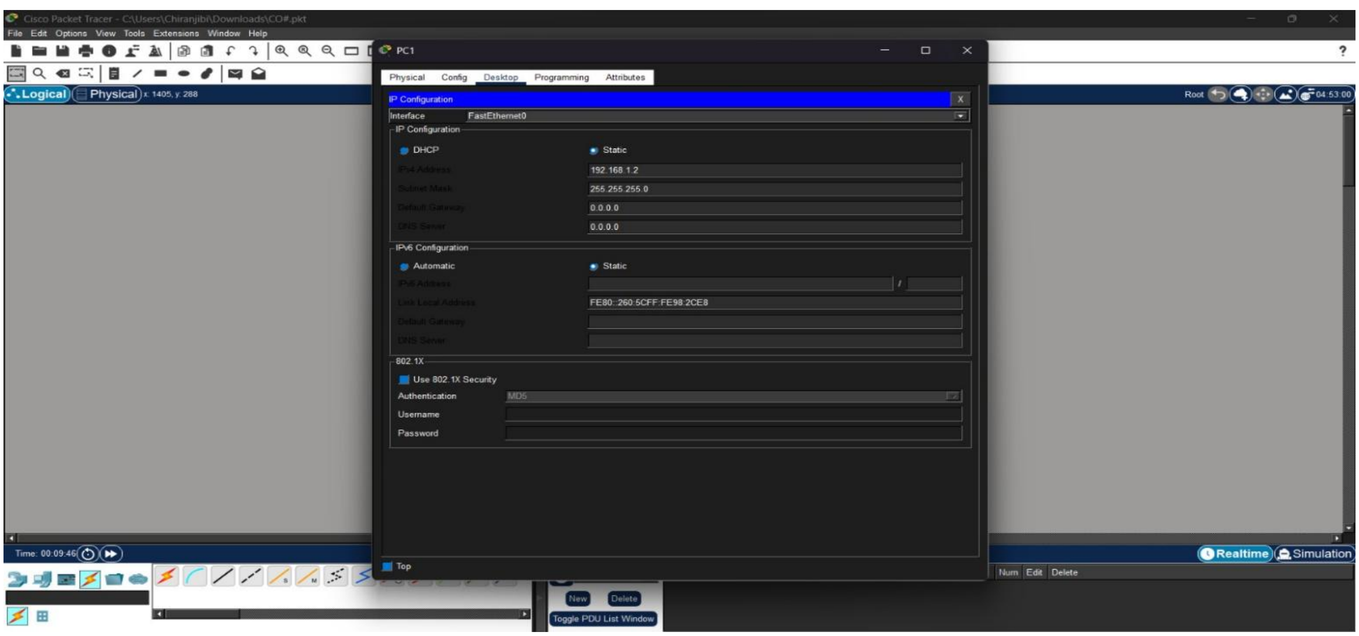
Step 3: Configure PC0's IP Address



Explanation:

In PC0's Desktop tab, the IP Configuration utility was opened and the IPv4 address was manually set to 192.168.1.1 with a subnet mask of 255.255.255.0. This gives PC0 a valid Network-layer address on the LAN.

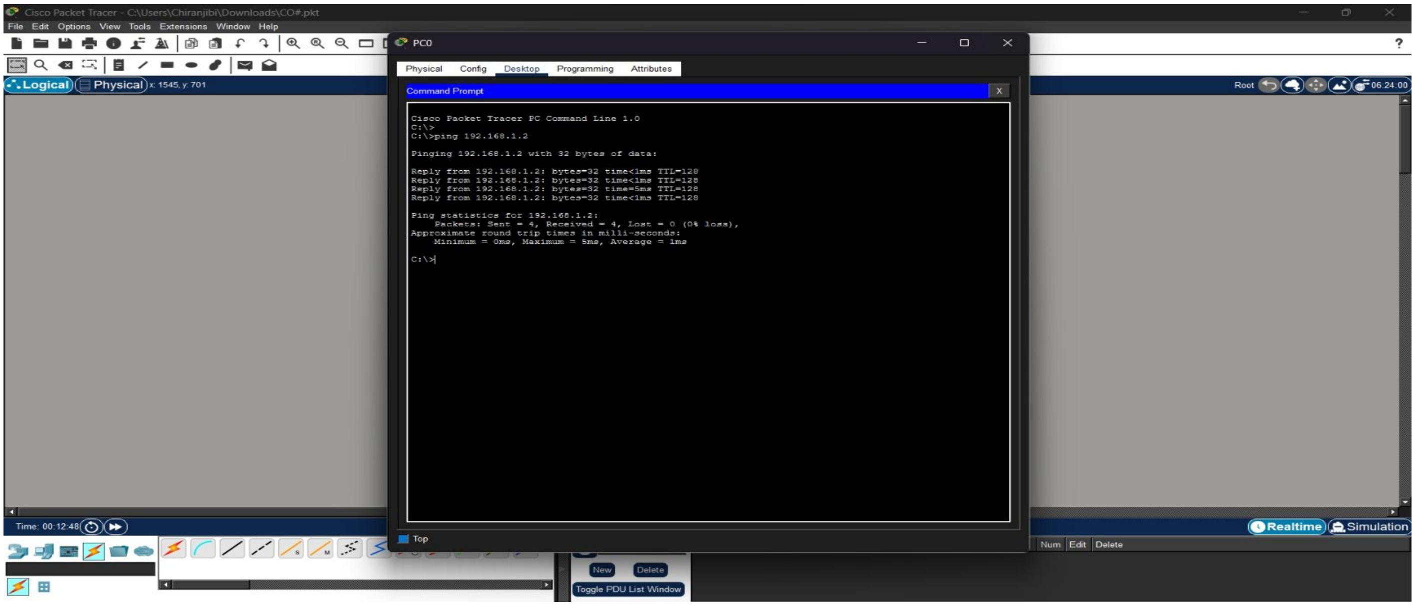
Step 4: Configure PC1's IP Address



Explanation:

The same procedure was repeated on PC1, assigning it the IP address 192.168.1.2 with subnet mask 255.255.255.0, placing both hosts in the same 192.168.1.0/24 subnet so they can communicate directly.

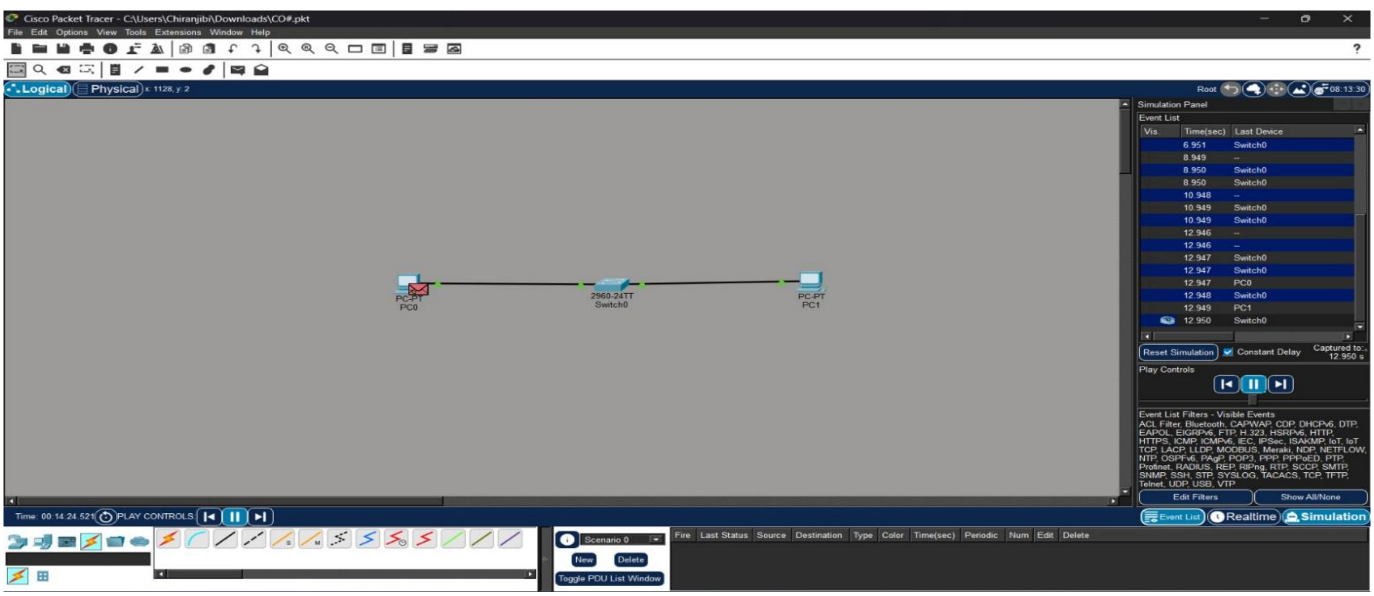
Step 5: Verify Connectivity with Ping



Explanation:

From PC0's Command Prompt, the command ping 192.168.1.2 was executed. The four successful Reply from 192.168.1.2: bytes=32 time<1ms TTL=128 messages confirm that IP connectivity between the two hosts has been established correctly.

Step 6: Switch to Simulation Mode and Apply Filters



Explanation:

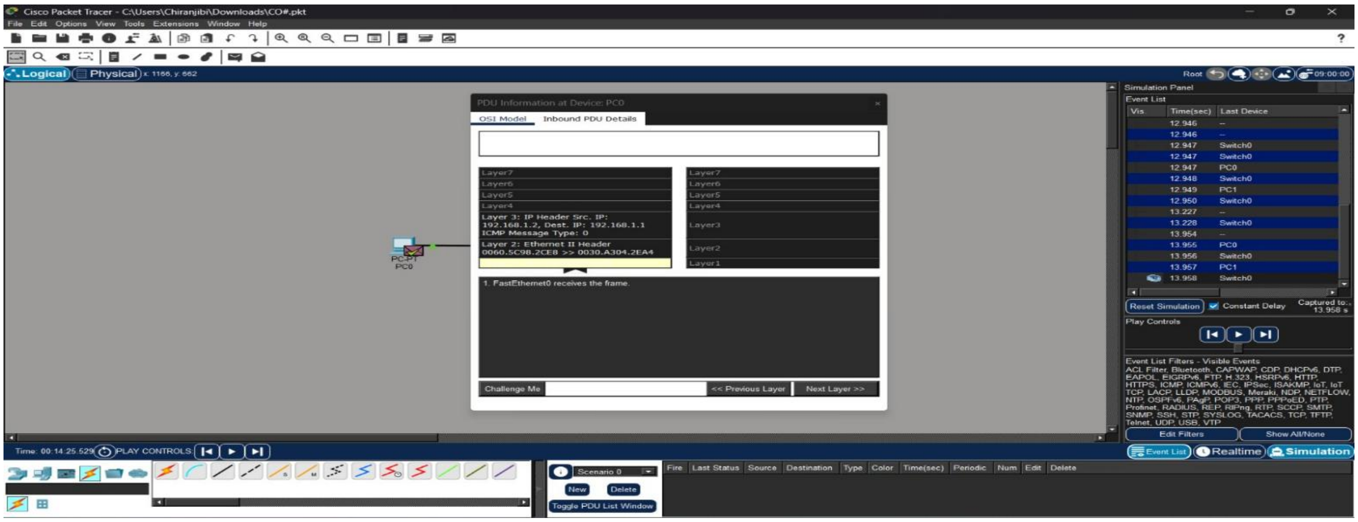
Packet Tracer was switched from Realtime mode to Simulation mode using the toggle at the bottom-right of the window. The Edit Filters option was used to select only ICMP, so that the event list would show just the ping-related traffic and not be cluttered with other background protocols.

Step 7: Capture the Ping Exchange

Explanation:

The ping command was re-issued on PC0, and the Auto Capture / Play button was pressed. The event list shows the ICMP Echo Request travelling from PC0 to the switch and on to PC1, followed by the ICMP Echo Reply travelling back, exactly mirroring what a live Wireshark capture on the link would show.

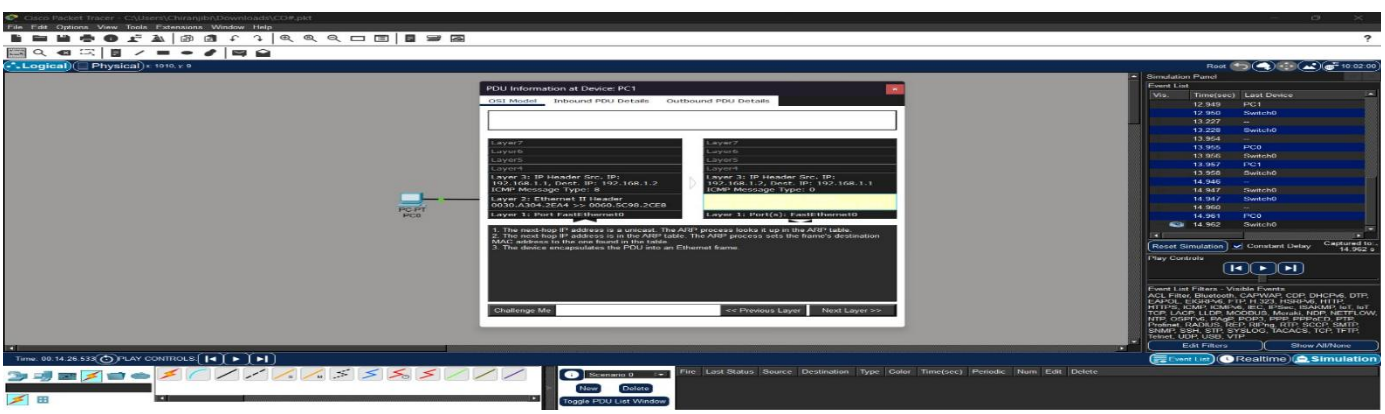
Step 8: Open the PDU and View the Ethernet II Header



Explanation:

Clicking the small envelope icon on the PC0–Switch link opened the PDU Information window. Under the OSI Model tab, Layer 2 (In Layers) displays the Ethernet II frame header: the Destination MAC Address (PC1's NIC address), the Source MAC Address (PC0's NIC address), and the Type field showing 0x0800, indicating that an IPv4 packet is encapsulated inside the frame.

Step 9: Note the Frame Check Sequence (FCS)



Explanation:

Scrolling to the bottom of the same Layer 2 view shows the Frame Check Sequence, a CRC value computed over the frame contents. This value lets the receiving NIC on PC1 verify that the frame was not corrupted during transmission, and to silently discard it if the check fails.

Step 10: Confirm Field Values in the Reply Direction

Explanation:

Opening the PDU for the Echo Reply frame travelling from PC1 back to PC0 shows the Source and Destination MAC addresses have swapped compared to the request — PC1's MAC is now the Source and PC0's MAC is now the Destination — confirming that each direction of an Ethernet conversation is addressed independently at Layer 2.

Explanation — Role of the Data Link Layer

The captured frames show that although PC0 and PC1 communicate using IP addresses at the Network layer, the actual delivery within the LAN happens purely through MAC addressing at the Data Link layer. When PC0 sends data to PC1, it encapsulates the IP packet inside an Ethernet frame with PC1's MAC address in the Destination field. The switch reads only this Destination MAC Address and forwards the frame out of the correct port using its MAC address (CAM) table, without ever inspecting the IP header. The EtherType field tells the receiving NIC which upper-layer protocol to hand the payload to, and the FCS lets it detect and discard corrupted frames. This confirms that reliable local delivery within a LAN segment is entirely a Layer-2 responsibility, working underneath and independently of Layer-3 IP addressing.

RESULT

A simple LAN with two PCs connected through a switch was successfully designed in Cisco Packet Tracer. IP connectivity was verified using the ping command, and the exchanged Ethernet frames were captured and analyzed. The Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS) fields were successfully identified, and the role of the Data Link layer in ensuring successful local frame delivery through MAC addressing was understood.

EXPERIMENT 3

QUESTION 3

Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

ANSWER

Aim

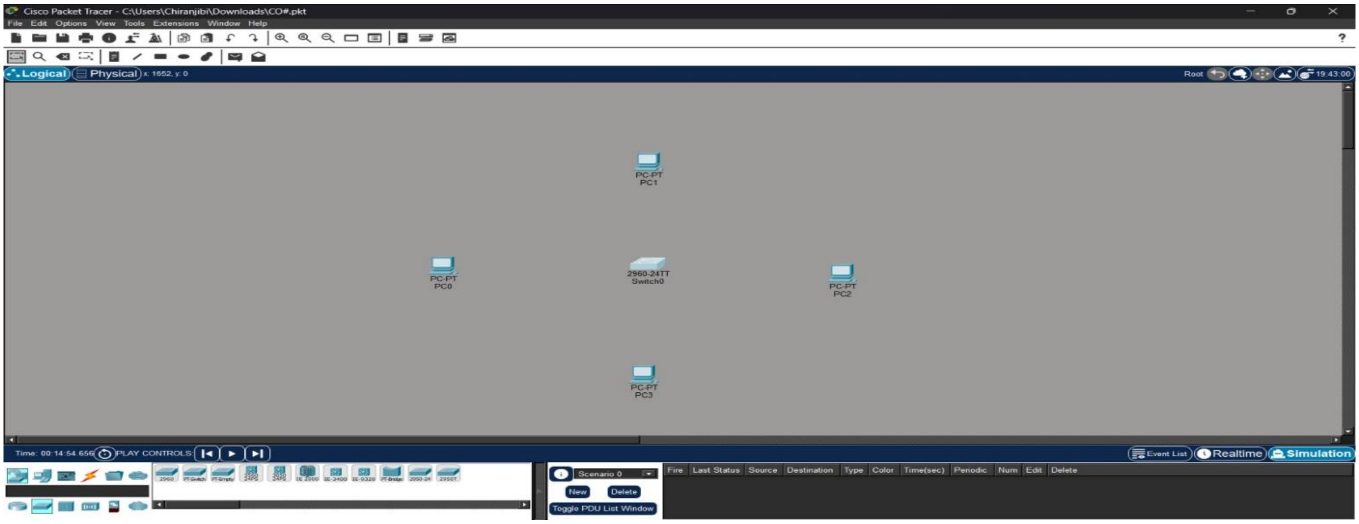
To construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer, clear the ARP cache of a host, initiate communication using the ping command, capture the resulting ARP Request and ARP Reply packets, and analyze the sender/target MAC and IP addresses to understand how ARP resolves logical IP addresses into physical MAC addresses before data transmission.

Procedure

1. Open Cisco Packet Tracer and create a new blank project.
2. Place four End Devices (PC0, PC1, PC2, PC3) and one 2960 Switch onto the workspace.
3. Connect all four PCs to the switch using Copper Straight-Through cables and confirm the link lights turn green.
4. Configure a unique IPv4 address on each PC within the same subnet (e.g., 192.168.1.1–192.168.1.4 / 255.255.255.0).
5. Open the Command Prompt on PC0 and view its current ARP cache using arp -a.
6. Clear PC0's ARP cache entry using arp -d (or power-cycle the PC) so that a fresh resolution must occur.
7. Switch Packet Tracer to Simulation Mode and set the Edit Filters to capture ARP and ICMP traffic.
8. From PC0, ping another host (e.g., PC2 at 192.168.1.3) and use Auto Capture / Play to capture the resulting traffic.
9. Open the PDU Information window for the broadcast ARP Request frame and record its Sender/Target MAC and IP fields.
10. Open the PDU Information window for the unicast ARP Reply frame and record its Sender/Target MAC and IP fields, then verify the updated ARP cache using arp -a.

Step-by-Step Implementation

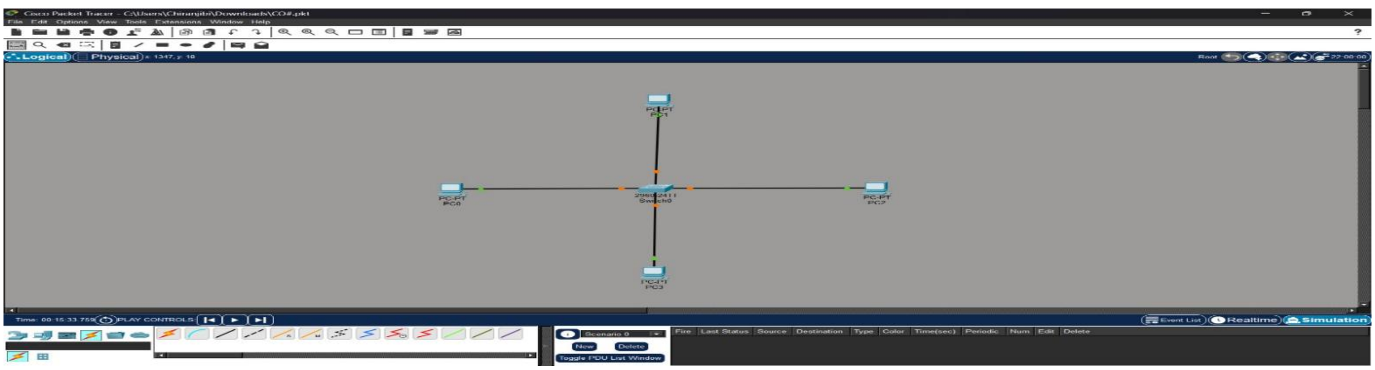
Step 1: Create the Topology



Explanation:

Four End Devices (PC0, PC1, PC2, PC3) and one 2960 Switch were placed on the workspace to form a multi-host LAN, as required to demonstrate that an ARP broadcast reaches every connected host.

Step 2: Cable the Devices



Explanation:

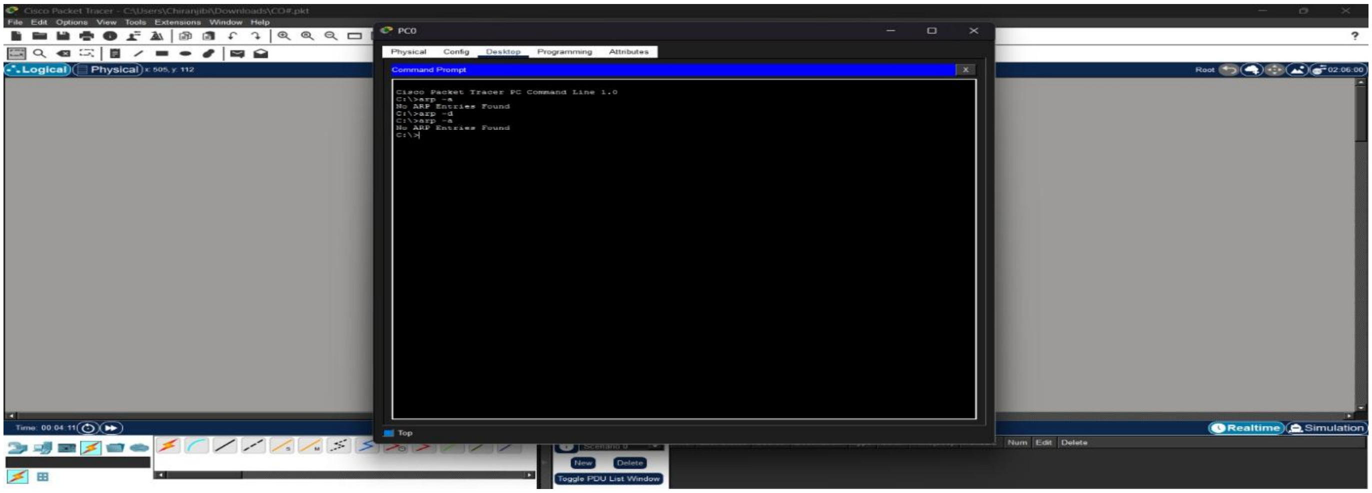
Each PC's FastEthernet0 port was connected to a separate FastEthernet port on the switch (0/1 to 0/4) using Copper Straight-Through cables. The green link lights on every connection confirm that all four hosts are physically attached to the same switch.

Step 3: Configure IP Addresses on All Hosts

Explanation:

Each PC was assigned a unique IPv4 address in the 192.168.1.0/24 network through Desktop → IP Configuration, ensuring all four hosts can reach each other directly at the Network layer once their MAC addresses are resolved.

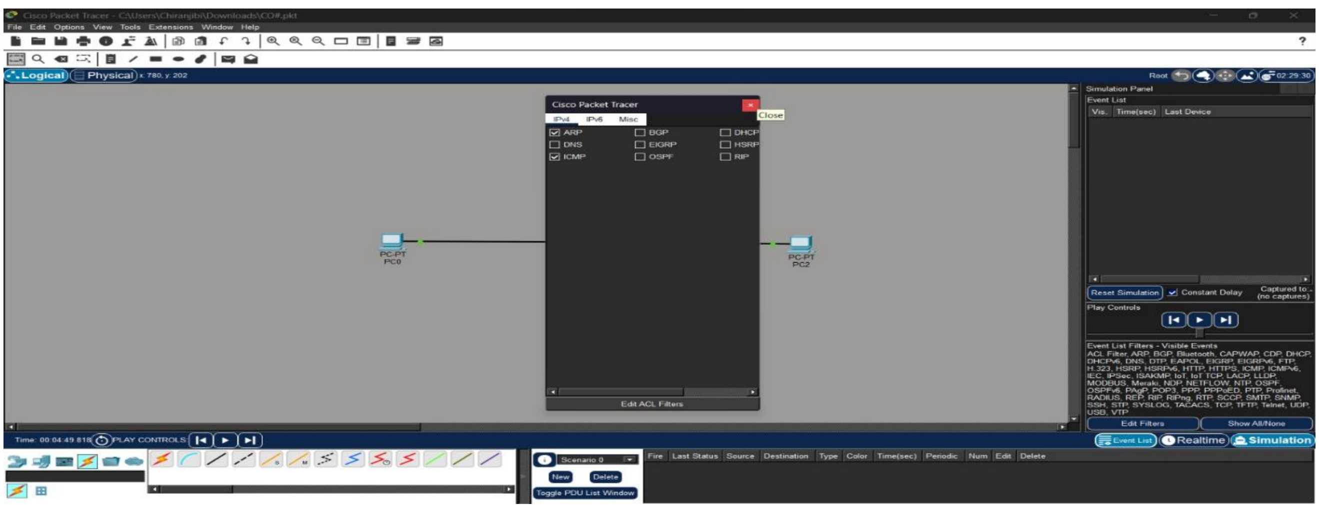
Step 4: View and Clear PC0's ARP Cache



Explanation:

On PC0's Command Prompt, arp -a was run first to display any existing IP-to-MAC mappings. The command arp -d was then used to delete the cached entry (where unsupported, PC0 was power-cycled from the Physical tab instead), guaranteeing that the next communication attempt would require a fresh ARP resolution.

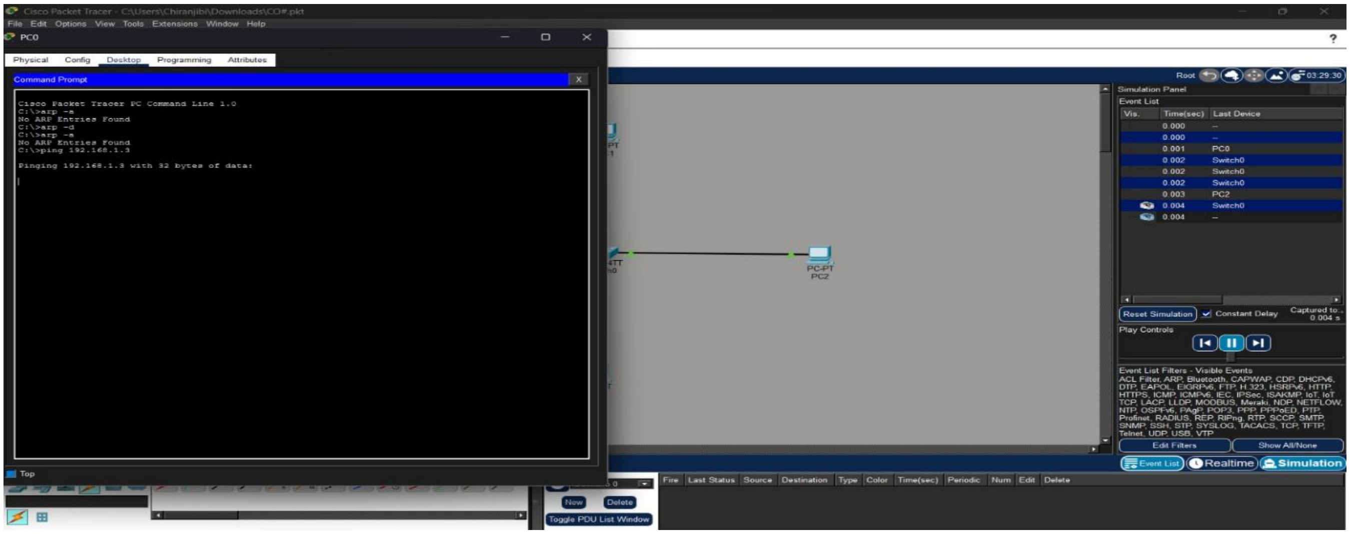
Step 5: Switch to Simulation Mode and Apply Filters



Explanation:

Packet Tracer was switched to Simulation mode, and the Edit Filters option was used to display only ARP and ICMP events, so the resolution process could be followed clearly without unrelated background traffic.

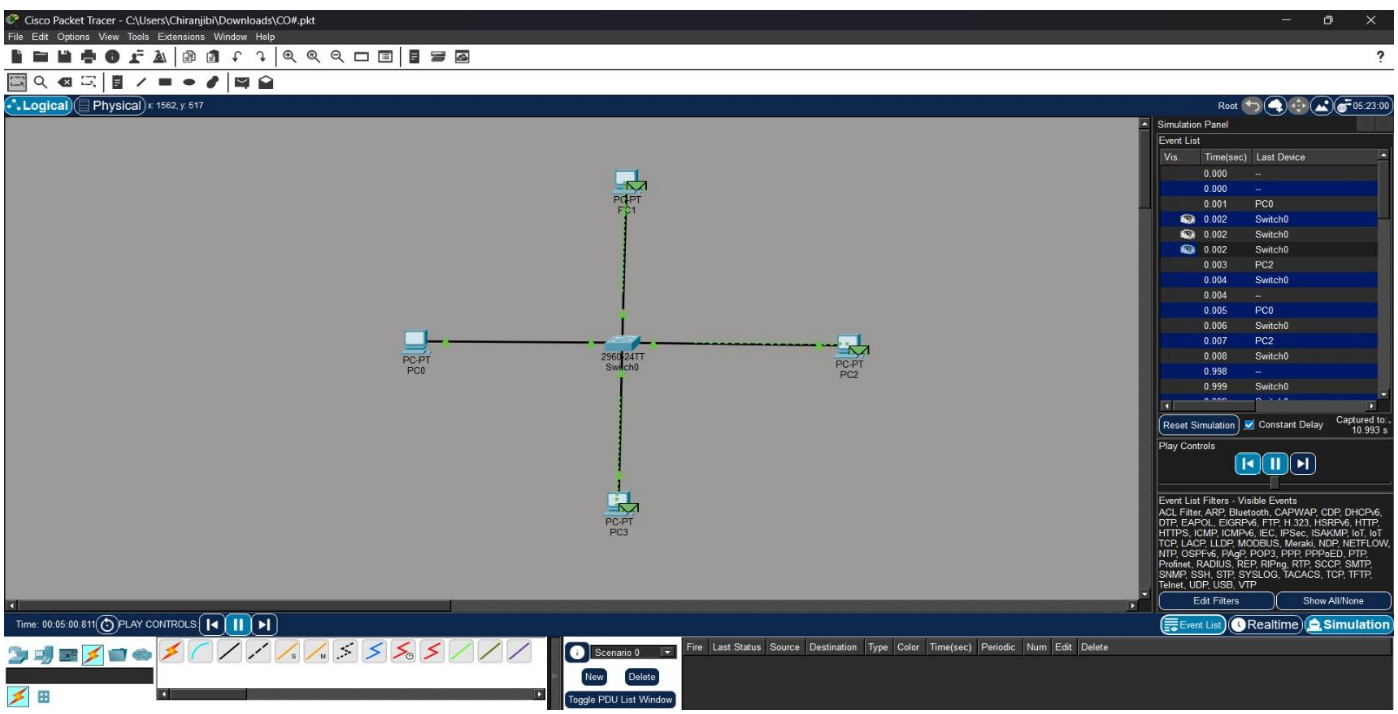
Step 6: Initiate the Ping and Capture Traffic



Explanation:

From PC0's Command Prompt, ping 192.168.1.3 was executed to contact PC2. Because PC0's ARP cache was empty, the very first event generated was a broadcast ARP packet rather than an ICMP packet, confirming that MAC-address resolution must happen before any data can be sent.

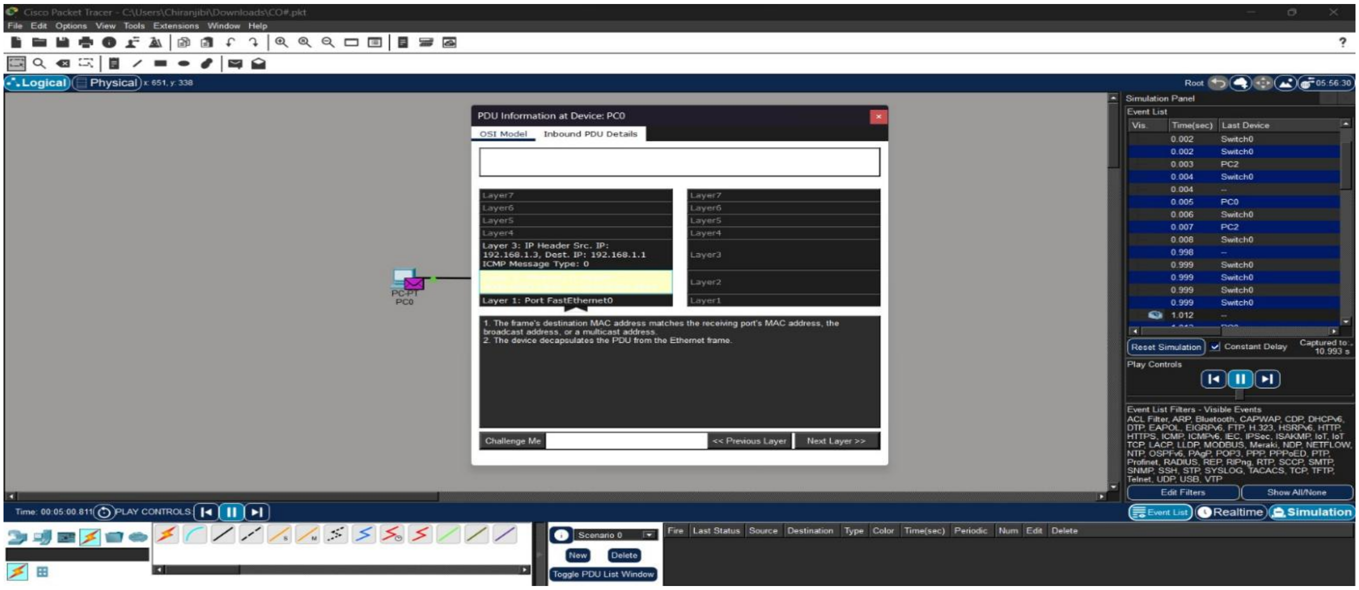
Step 7: Observe the ARP Request Being Flooded



Explanation:

The event list shows the switch receiving the broadcast ARP Request from PC0 and flooding a copy out of every port except the one it arrived on, so PC1, PC2, and PC3 all receive the request even though it is intended for only one of them.

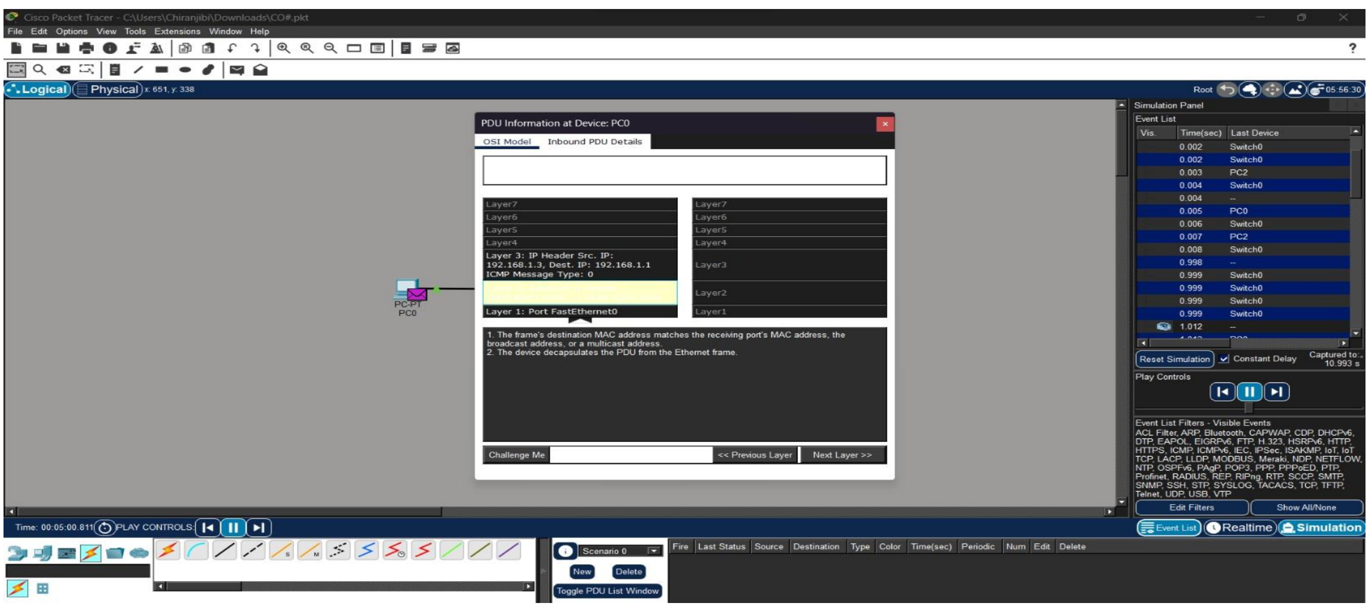
Step 8: Inspect the ARP Request PDU



Explanation:

Opening the PDU for the frame leaving PC0 shows the ARP Request fields: Opcode = 1 (Request), Sender MAC/IP set to PC0's own address (192.168.1.1), and Target MAC left as 00:00:00:00:00:00 since it is still unknown, while Target IP is set to 192.168.1.3, the address being resolved.

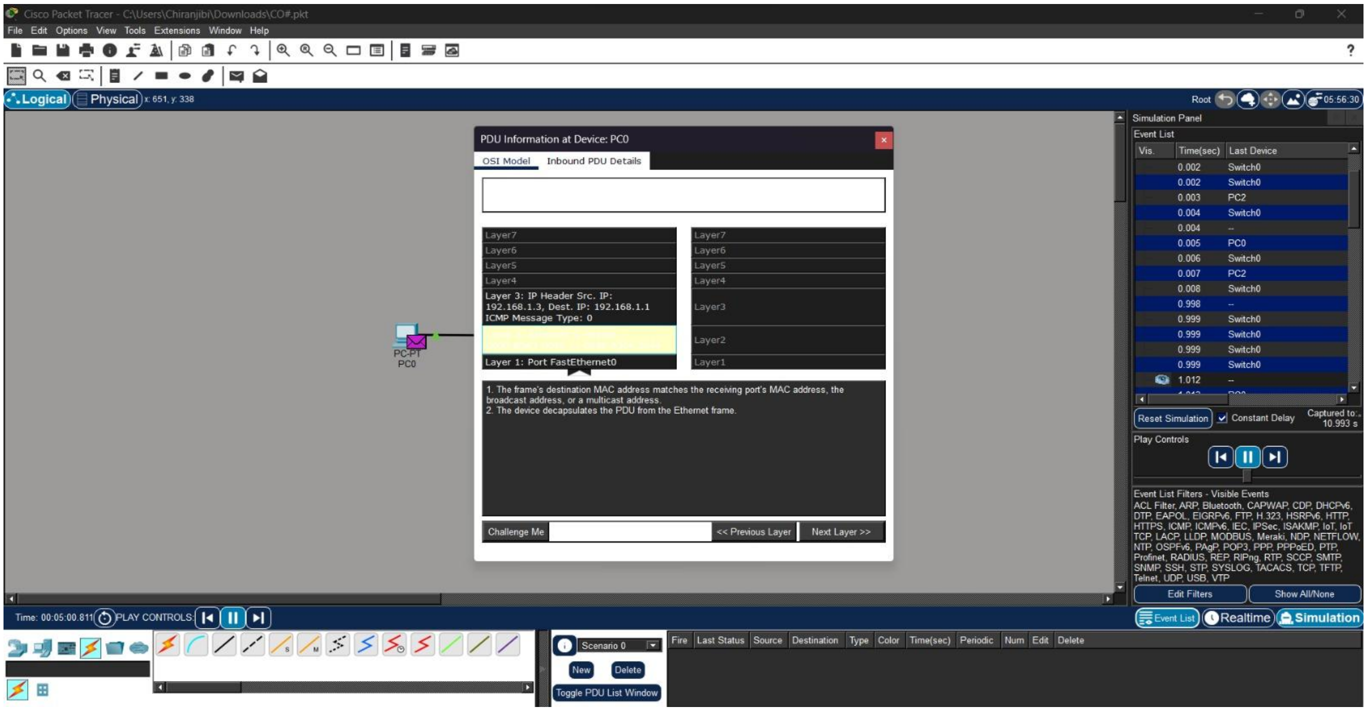
Step 9: Inspect the ARP Reply PDU



Explanation:

Only PC2 recognised its own IP address in the Target IP field and responded with a unicast ARP Reply. Its PDU shows Opcode = 2 (Reply), Sender MAC/IP now set to PC2's own address (192.168.1.3), and Target MAC/IP set back to PC0's address (192.168.1.1), completing the resolution.

Step 10: Verify the Updated ARP Cache and Subsequent ICMP Traffic



Explanation:

Running arp -a on PC0 again shows a new entry mapping 192.168.1.3 to PC2's MAC address. The Simulation event list also shows that the ICMP Echo Request/Reply frames that follow are sent directly using this resolved MAC address, with no further ARP broadcasts needed for the rest of the session.

Explanation — Role of ARP

Clearing PC0's ARP cache forced it to broadcast an ARP Request to every host on the LAN before it could address the ICMP Echo Request to PC2. Because the request was broadcast (Destination MAC = FF:FF:FF:FF:FF:FF), the switch flooded it to every connected port, and every host received a copy — yet only PC2, whose IP address matched the Target IP field, replied. This shows that ARP resolution is IP-address specific even though the query itself is broadcast at Layer 2. The unicast ARP Reply then let PC0 populate its ARP cache with PC2's MAC address, after which all further packets to 192.168.1.3 during the same session used the resolved MAC address directly instead of broadcasting again. This demonstrates ARP's essential role as the glue between the logical (IP) addressing used by the Network layer and the physical (MAC) addressing used by the Data Link layer, allowing data transmission to proceed only once the destination's physical address is known.

RESULT

A multi-host LAN was successfully constructed in Cisco Packet Tracer, the ARP cache of a host was cleared, and the resulting ARP Request and ARP Reply packets were captured and analyzed. The Sender/Target MAC and IP address fields were identified for both packets, and the role of ARP in resolving IP addresses to MAC addresses prior to data transmission was verified and understood.